StickShield is a flexible acoustic metamaterial patch that can be directly attached to the surface of a microphone. It effectively defends against adversarial audio attacks, inaudible ultrasonic attacks, and laser injection attacks by absorbing acoustic energy within specific frequency bands, while preserving the intelligibility of normal voice commands. This artifact includes 3D printing model files (in .SLDPRT and .STL formats) and demonstration videos (in .mp4 format). The former are used to fabricate the metamaterial units, and the latter are used to showcase StickShield's defensive effectiveness against typical inaudible and adversarial attacks.

The StickShield unit consists of two structural layers. The top layer is a membrane made of silicone rubber, with a Young's modulus of 9×10^6 Pa and a density of 1200 kg/m³, primarily used to reflect and absorb high-frequency ultrasonic signals. The bottom layer is a box-like cavity structure made of soft polyurethane rubber, with a Young's modulus of 6.5×10^6 Pa and a density of 1040 kg/m³, used to absorb adversarial perturbations in the audible frequency band. Regarding delivery, the .SLDPRT files require SolidWorks to open, allowing users to adjust the unit's geometric parameters according to the actual microphone dimensions of the target device; the .STL files are in standard 3D printing format and can be directly used to print individual units, and multiple units can be combined via copy and paste to accommodate different microphone array layouts. After printing, the structure is attached to the device's microphone surface with the top side facing outward, using double-sided tape or other adhesives or fastening mechanisms.

For installation and fabrication, users first obtain the 3D blueprints from the open-science platform, then open the .SLDPRT files in SolidWorks to adjust the unit's length, width, and height according to the target microphone size. Next, slicing software is used to generate G-code, and the part is printed using an SLA or FDM 3D printer, with the top layer recommended to be printed in silicone rubber and the bottom layer in soft polyurethane rubber. Finally, support structures are removed and the surface is cleaned before attachment. This artifact does not rely on any datasets; attack tests use standard synthetic voice commands. The runtime environment consists of 3D printing equipment and video players for viewing demonstrations. An SLA 3D printer with a precision of at least 0.1 mm is recommended. The overall experimental workflow includes printing the unit, attaching it to the microphone, launching the attack, and recording the defense outcome, with printing taking about 2–4 hours and attachment and testing about 30 minutes.

In terms of the experimental workflow, users first attach the printed unit to the microphone surface, then use standard attack devices to inject attack commands into the microphone, and finally observe whether the voice assistant executes the attack commands to determine whether the defense is effective. Regarding expected defensive performance, after applying StickShield, the success rate of attack commands should drop significantly, with an attack failure rate above 92%, while the recognition accuracy for normal voice commands should remain above 95%. Detailed defense performance data can be found in Section 6 of the main paper. Users may also conduct custom experiments, such as modifying geometric parameters in the .SLDPRT files to fit different microphone sizes, constructing larger patches covering multiple microphone arrays by copying and pasting .STL models, or trying alternative flexible materials when printing conditions are limited, though they should be aware that material changes may affect acoustic performance. This work is intended solely for defensive security research and does not involve the collection of sensitive data from human subjects.